

SHISANJIANFANG, China

WMO: 514950

Lat: 43.2124N

Lon: 91.7424E

Elev: 732

StdP: 92.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.3	(c) -15.3	(d) -28.4	(e) 0.3	(f) -8.9	(g) -26.0	(h) 0.4	(i) -7.9	(j) 23.0	(k) -4.2	(l) 21.1	(m) -5.8	(n) 1.8	(o) 0	(p) 0.851

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 38.4	(d) 17.4	(e) 36.8	(f) 17.0	(g) 35.4	(h) 16.5	(i) 18.6	(j) 33.3	(k) 18.0	(l) 32.7	(m) 17.4	(n) 32.1	(o) 6.2	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	10.9	24.0	12.5	9.8	24.7	11.0	8.9	25.3	55.9	32.9	53.6	32.6	51.9	32.0	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.3	(b) 21.9	(c) 19.8	(e) -18.6	(f) 40.9	(g) 2.7	(h) 1.4	(i) -20.6	(j) 41.9	(k) -22.1	(l) 42.8	(m) -23.6	(n) 43.5	(o) -25.6	(p) 44.6		
(a) 24.3	(b) 21.9	(c) 19.8	(e) -19.3	(f) 19.8	(g) 2.5	(h) 1.3	(i) -21.1	(j) 20.7	(k) -22.6	(l) 21.4	(m) -24.0	(n) 22.2	(o) -25.9	(p) 23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-7.9	-1.6	6.7	14.8	21.2	27.0	29.1	27.8	21.6	12.8	3.2	-5.9
	DBStd	13.50	3.37	4.09	5.62	5.38	4.48	3.78	3.47	3.62	4.26	4.76	4.78	3.94
	HDD10.0	1761	555	324	132	21	1	0	0	0	1	28	209	491
	HDD18.3	3286	813	558	361	127	24	1	0	1	18	179	455	749
	CDD10.0	2665	0	0	30	165	348	510	592	552	350	114	4	0
	CDD18.3	1147	0	0	0	22	113	261	333	294	117	7	0	0
	CDH23.3	13808	0	0	1	185	1103	3218	4431	3738	1088	44	0	0
	CDH26.7	7041	0	0	0	38	426	1656	2508	2009	399	4	0	0
Wind	WSAvg	7.7	3.9	6.2	8.1	9.4	9.9	10.2	9.9	9.4	8.2	7.0	5.7	4.2
Precipitation	PrecAvg	19	1	0	0	1	1	5	5	4	2	0	1	1
	PrecMax	38	4	1	0	12	5	17	23	21	11	2	5	2
	PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	10	1	0	0	3	2	5	7	6	3	1	2	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	11.2	22.6	30.2	35.1	39.1	41.0	39.5	34.6	27.1	17.2	5.1
		MCWB	-1.7	2.9	8.8	13.2	15.0	17.2	18.6	17.7	15.6	12.2	7.1	-0.1
	2%	DB	0.3	8.8	20.0	27.3	33.0	36.9	38.9	37.7	32.6	24.4	14.0	3.0
		MCWB	-3.6	1.5	7.6	11.5	14.3	16.6	17.9	17.0	14.8	10.7	5.5	-1.4
	5%	DB	-1.2	6.9	17.7	25.4	31.1	35.4	37.3	36.1	30.9	22.4	12.1	1.3
		MCWB	-4.7	0.3	6.7	10.7	13.7	16.2	17.4	16.7	14.2	9.9	4.5	-2.7
	10%	DB	-2.7	5.2	15.4	23.3	29.1	33.8	35.7	34.5	29.0	20.5	10.2	-0.1
		MCWB	-5.8	-0.6	5.6	9.8	12.9	15.8	17.0	16.2	13.5	9.0	3.6	-3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	3.2	9.3	13.6	16.0	18.6	19.9	19.3	16.6	12.6	7.5	0.3
		MCDB	2.3	10.5	20.6	28.0	33.3	32.4	36.3	32.9	31.9	25.8	15.8	4.5
	2%	WB	-3.4	1.7	8.2	12.3	15.0	17.8	18.9	18.2	15.5	11.4	5.9	-1.2
		MCDB	0.0	8.2	18.9	25.5	30.9	32.3	34.0	33.4	30.1	23.2	13.2	2.9
	5%	WB	-4.6	0.6	7.1	11.3	14.2	17.2	18.3	17.5	14.7	10.4	4.8	-2.5
		MCDB	-1.4	6.7	16.7	23.6	28.9	31.7	32.9	32.9	29.1	21.4	11.5	1.2
	10%	WB	-5.7	-0.6	5.9	10.4	13.4	16.6	17.7	16.9	13.9	9.3	3.7	-3.6
		MCDB	-2.8	5.0	15.0	22.2	27.7	31.1	32.1	32.1	28.2	19.7	10.0	-0.3
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	10.6	10.7	10.2	9.7	9.6	10.1	10.6	10.4	8.8	7.9
		MCDBR	9.4	11.3	12.3	12.6	12.1	10.9	11.0	11.7	12.0	12.1	11.0	8.5
		MCWBR	6.8	7.1	6.1	5.8	5.1	3.9	3.8	4.2	4.8	5.7	6.3	6.0
	5% WB	MCDBR	9.0	10.8	11.2	11.2	10.5	9.5	9.5	10.1	10.9	11.1	10.3	8.4
		MCWBR	6.7	6.9	5.7	5.2	4.6	3.4	3.2	3.6	4.5	5.2	6.0	6.0
Clear-Sky Solar Irradiance	taub	0.336	0.373	0.416	0.461	0.463	0.469	0.458	0.432	0.413	0.396	0.363	0.334	
	taud	2.120	2.061	1.968	1.916	1.966	2.029	2.102	2.155	2.126	2.081	2.112	2.130	
	Ebn at Noon	760	795	808	804	815	810	814	823	805	757	717	719	
	Edh at Noon	101	128	160	182	179	168	155	142	135	124	102	92	
All-Sky Solar Radiation	RadAvg	2.19	3.24	4.67	5.92	6.98	7.11	6.76	6.28	5.34	3.86	2.44	1.86	
	RadStd	0.13	0.16	0.18	0.35	0.30	0.45	0.27	0.31	0.20	0.13	0.14	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.19	+2.01	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.56	N/A	N/A	+0.54	N/A	-0.78	N/A	N/A	+146	+101

Nomenclature: See separate page